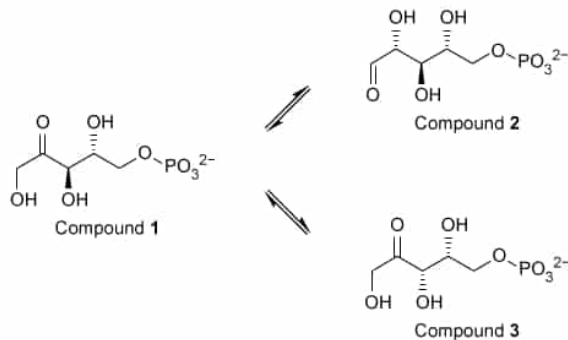


At the end of the oxidative phase of the pentose phosphate pathway, Compound 1 is isomerized into either Compound 2 or Compound 3, and all three compounds are maintained at equilibrium ratios.



Compounds 1, 2, and 3 can be best described as phosphate derivatives of:

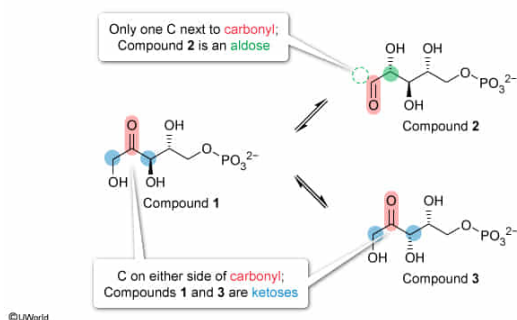
- ☐ A. a ketose, an aldose, and an aldose, respectively. [2%]
- ☐ B. an aldose, a ketose, and an aldose, respectively. [14%]
- ☐ C. an aldose, a ketose, and a ketose, respectively. [2%]
- ☒ D. a ketose, an aldose, and a ketose, respectively. [80%]

Correct

79% answered correctly

Explanation:

Aldoses contain aldehydes and ketoses contain ketones, respectively



Monosaccharides are the simplest carbohydrates and have the general formula $(\text{CH}_2\text{O})_n$. Consequently, monosaccharides have **one degree of unsaturation**, which comes in the form of a **carbonyl group** (for linear monosaccharides) or a ring structure (through **intramolecular addition** to the **carbonyl group**). The position of the carbonyl carbon—also called the **anomeric carbon** in sugars—is used to divide monosaccharides and their derivatives into one of two classes: **ketoses** and **aldoses**.

In linear ketoses, the anomeric carbon is a **ketone**, which is attached to **two carbon atoms** in addition to its double bond to oxygen. Cyclization converts the ketose anomeric carbon to a **hemiketal**. Conversely, in linear aldoses the anomeric carbon is an **aldehyde**, which is attached to **one hydrogen atom**, in addition to a carbon and the double bond to oxygen. Cyclization converts the aldose anomeric carbon to a

hemiacetal.

In this question, all three intermediates of the **pentose phosphate pathway** are represented in their linear form. **Compounds 1 and 3** (ie, ribulose 5-phosphate and xylulose 5-phosphate, respectively) have ketones on their C2 atom; therefore, **Compounds 1 and 3 are ketoses**. In contrast, **Compound 2** (ribose 5-phosphate) has an aldehyde for its C1 atom; therefore, **Compound 2 is an aldose**. Listed in numerical order, Compounds **1, 2, and 3** are a ketose, an aldose, and a ketose, respectively.

(Choice A) The carbonyl carbon of Compound 3 is attached to two other carbons. Therefore, the carbonyl carbon of Compound 3 is a ketone and Compound 3 is a ketose, not an aldose.

(Choices B and C) The carbonyl carbon of Compound 1 is attached to two other carbons. Therefore, the carbonyl carbon of Compound 1 is a ketone, and Compound 1 is a ketose. The carbonyl carbon of Compound 2 is attached to one hydrogen, as well as to another carbon. Therefore, the carbonyl carbon of Compound 2 is an aldehyde and Compound 2 is an aldose, not a ketose.

Educational objective:

Monosaccharides can be classified as either ketoses or aldoses, based on the intramolecular location of their anomeric carbon. For linear monosaccharides, the anomeric carbon of aldoses is an aldehyde, whereas the anomeric carbon of ketoses is a ketone.

Time spent:0 sec

Subject:
Biochemistry

QID:403970

System:
Carbohydrates, Nucleotides, and Lipids